BOSTON UNIVERSITY

FACULTY OF COMPUTING AND DATA SCIENCE

Thesis

WHAT DOES A PHYSICS NETWORK ACTUALLY LEARN? MECHANISTIC INTERPRETABILITY ACROSS A SPECTRUM OF PHYSICS-INFORMED NEURAL ARCHITECTURES

by

ACHUTHAN RATHINAM

B.S., **[TODO: Institution]**, **[TODO: Year]**

Submitted in partial fulfillment of the

requirements for the degree of

Master of Science

2026

© 2026 by
ACHUTHAN RATHINAM
All rights reserved

Approved by

First Reader

Siddharth Mishra-Sharma, Ph.D.
Assistant Professor of Computing & Data Sciences

Second Reader

Naomi Saphra, Ph.D.
Assistant Professor of Computing & Data Sciences

ACKNOWLEDGMENTS

[Advisor, readers, collaborators, compute resources, funding.]

WHAT DOES A PHYSICS NETWORK ACTUALLY LEARN?

MECHANISTIC INTERPRETABILITY ACROSS A

SPECTRUM OF PHYSICS-INFORMED NEURAL

ARCHITECTURES

ACHUTHAN RATHINAM

ABSTRACT

[Write last. One paragraph each on the problem, the method, the results, and the conclusions.]

TABLE OF CONTENTS

ACKNOWLEDGMENTS iv

ABSTRACT v

TABLE OF CONTENTS vi

LIST OF TABLES xi

LIST OF FIGURES xii

LIST OF ABBREVIATIONS xiii

CHAPTER ONE: INTRODUCTION 1

- Motivation 1
- The Central Question 1
- Why Coulomb Dynamics 1
- Thesis Statement 1
- Research Questions 1
- Contributions 1
- Thesis Organization 1

CHAPTER TWO: BACKGROUND AND RELATED WORK 2

- Learning Physical Dynamics with Neural Networks 3
- A Spectrum of Physics Priors 3
- Symmetry-Constrained Architectures 3
  - Equivariance 3
  - Equivariant Graph Neural Networks 3
  - Attention-Based Equivariant Message Passing 3
  - Equivariant Transformers 3

Physics-Explicit Architectures 3
Hamiltonian Neural Networks 3
Hamiltonian Networks at Multi-Particle Scale 3
Unconstrained Baselines 3
Mechanistic Interpretability 3
Linear Probing and Its Pitfalls 3
Causal Intervention 3
Representational Similarity 3
Grokking and Delayed Generalization 3
Interpretability of Physics and Scientific ML Models 3
Summary and Gap 3
CHAPTER THREE: THE COULOMB DYNAMICS TESTBED 4
Physical System 4
Simulator 4
Physics Verification 4
Prediction Task 4
Dataset 4
Ground-Truth Physical Quantities 4
CHAPTER FOUR: ARCHITECTURES AND TRAINING 5
Architecture Suite 5
Implementation Details 5
Non-Equivariant Graph Neural Network 5
EGNN 5
Attention EGNN 5

Equivariant Transformer 5
Hamiltonian Neural Network 5
Matched Training Protocol 5
Output Accuracy 5
Equivariance Checks 5
CHAPTER FIVE: INTERPRETABILITY METHODOLOGY 6
Activation Capture 6
Linear Probing 6
Probe Family and Metrics 6
Control Baselines 6
Causal Ablation 6
Activation Patching 6
Representational Similarity 6
Training-Dynamics Tracking 6
Compute and Reproducibility 6
CHAPTER SIX: RESULTS I: WHAT IS ENCODED 7
Physics Encoding Profiles 7
Geometry Versus Law 7
Depth of Emergence 7
The Hamiltonian Network Versus the Rest 7
Unconstrained Baselines 7
Summary of RQ1 Findings 7
CHAPTER SEVEN: RESULTS II: WHAT IS USED 8
Ablation Results 8

Correlate Versus Cause 8
Activation Patching: Tracing a Charge Flip 8
Encodes Versus Uses 8
Summary of RQ2 Findings 8
CHAPTER EIGHT: RESULTS III: REPRESENTATIONAL GEOMETRY AND DYNAMICS 9
Cross-Architecture Similarity 9
Does the Implementation of Symmetry Matter? 9
Physics-Explicit Versus Symmetry-Constrained 9
Sensitivity to the Matched-Depth Definition 9
Generalization Dynamics 9
Summary of RQ3–RQ5 Findings 9
CHAPTER NINE: DISCUSSION 10
Revisiting the Central Hypothesis 10
Implications for Physics-Informed Machine Learning 10
Implications for Interpretability 10
Learned Law Versus Shortcut, Revisited 10
Limitations 10
CHAPTER TEN: CONCLUSION AND FUTURE WORK 11
Summary of Contributions 11
Future Work 11
APPENDIX A: SIMULATOR AND VERIFICATION DETAILS 12
APPENDIX B: HYPERPARAMETERS AND TRAINING CURVES 13
APPENDIX C: PROBE CONTROL BASELINES AND EXTENDED RESULTS 14

APPENDIX D: ADDITIONAL REPRESENTATIONAL SIMILARITY ANALYSES 15

APPENDIX E: CODE AND DATA AVAILABILITY 16

BIBLIOGRAPHY 17

CURRICULUM VITAE 18

LIST OF TABLES

[Fills in automatically once tables exist.]

LIST OF FIGURES

[Fills in automatically once figures exist.]

LIST OF ABBREVIATIONS

CKA.....Centered Kernel Alignment

EGNN.....E(n) Equivariant Graph Neural Network

GNN.....Graph Neural Network

HNN.....Hamiltonian Neural Network

MLP.....Multilayer Perceptron

MSE.....Mean Squared Error

RQ.....Research Question

SO(3).....Special Orthogonal Group in Three Dimensions (rotations)

CHAPTER ONE

INTRODUCTION

[Motivates the problem, poses the central question, states the hypothesis and the five research questions, and lists the intended contributions. Adapted from the proposal's Problem Statement and Proposed Research; Contributions and Organization are written last.]

Motivation

The Central Question

Why Coulomb Dynamics

Thesis Statement

Research Questions

Contributions

Thesis Organization

CHAPTER TWO

BACKGROUND AND RELATED WORK

[Everything the reader needs to follow Chapters 3–8: the spectrum of physics priors, each architecture family, the interpretability tools, and the closest prior work (Cranmer et al. 2020 in particular). Related-work survey still in progress.]

Learning Physical Dynamics with Neural Networks

A Spectrum of Physics Priors

Symmetry-Constrained Architectures

Equivariance

Equivariant Graph Neural Networks

Attention-Based Equivariant Message Passing

Equivariant Transformers

Physics-Explicit Architectures

Hamiltonian Neural Networks

Hamiltonian Networks at Multi-Particle Scale

Unconstrained Baselines

Mechanistic Interpretability

Linear Probing and Its Pitfalls

Causal Intervention

Representational Similarity

Grokking and Delayed Generalization

Interpretability of Physics and Scientific ML Models

Summary and Gap

CHAPTER THREE

THE COULOMB DYNAMICS TESTBED

[Fully specifies the physical system, simulator, verification checks, prediction task, dataset, and the analytic ground-truth quantities that serve as probe targets (the "answer key"). The task is multi-step position prediction.]

Physical System

Simulator

Physics Verification

Prediction Task

Dataset

Ground-Truth Physical Quantities

CHAPTER FOUR

ARCHITECTURES AND TRAINING

[Defines the architecture suite, per-model implementation details (including where "layer k activations" are read from), the matched training protocol, and shows comparable output accuracy across models so that later representational differences are not explained by one model being worse. Five architectures; the MLP is left out.]

Architecture Suite

Implementation Details

Non-Equivariant Graph Neural Network

EGNN

Attention EGNN

Equivariant Transformer

Hamiltonian Neural Network

Matched Training Protocol

Output Accuracy

Equivariance Checks

CHAPTER FIVE

INTERPRETABILITY METHODOLOGY

[Defines every measurement instrument once: activation capture, linear probing with control baselines, causal ablation, activation patching, CKA (including the chosen definition of "matched depth" across heterogeneous models), and training-dynamics tracking.]

Activation Capture

Linear Probing

Probe Family and Metrics

Control Baselines

Causal Ablation

Activation Patching

Representational Similarity

Training-Dynamics Tracking

Compute and Reproducibility

CHAPTER SIX

RESULTS I: WHAT IS ENCODED

[Answers RQ1. Layer-by-layer physics encoding profiles for every architecture and ground-truth quantity, always shown against the control baselines; compares geometry (distance, direction) with law (force magnitude, potential).]

Physics Encoding Profiles

Geometry Versus Law

Depth of Emergence

The Hamiltonian Network Versus the Rest

Unconstrained Baselines

Summary of RQ1 Findings

CHAPTER SEVEN

RESULTS II: WHAT IS USED

[Answers RQ2. Ablation of probed directions and activation patching of a charge flip, separating quantities the models merely encode from quantities they causally use.]

Ablation Results

Correlate Versus Cause

Activation Patching: Tracing a Charge Flip

Encodes Versus Uses

Summary of RQ2 Findings

CHAPTER EIGHT

RESULTS III: REPRESENTATIONAL GEOMETRY AND DYNAMICS

[Answers RQ3 and RQ4 with cross-architecture CKA (plus a sensitivity check on the matched-depth definition), and RQ5 with probe accuracy over training. RQ5 is exploratory and may be folded into the Discussion.]

Cross-Architecture Similarity

Does the Implementation of Symmetry Matter?

Physics-Explicit Versus Symmetry-Constrained

Sensitivity to the Matched-Depth Definition

Generalization Dynamics

Summary of RQ3–RQ5 Findings

CHAPTER NINE

DISCUSSION

[Revisits the hypothesis against the results, draws implications for physics ML (representational alignment as a third evaluation axis) and for interpretability (what analytic ground truth buys), and states limitations.]

Revisiting the Central Hypothesis

Implications for Physics-Informed Machine Learning

Implications for Interpretability

Learned Law Versus Shortcut, Revisited

Limitations

CHAPTER TEN

CONCLUSION AND FUTURE WORK

[Restates the contributions with results in hand and lists future directions (other force laws, nonlinear probes, symbolic distillation, larger systems, transfer to molecular models).]

Summary of Contributions

Future Work

APPENDIX A: SIMULATOR AND VERIFICATION DETAILS

[Integrator derivation, softening choice, full verification tables.]

APPENDIX B: HYPERPARAMETERS AND TRAINING CURVES

[Per-model hyperparameter tables and all training curves.]

APPENDIX C: PROBE CONTROL BASELINES AND EXTENDED RESULTS

[Control-baseline plots that do not fit in Chapter 6; per-seed variance.]

APPENDIX D: ADDITIONAL REPRESENTATIONAL SIMILARITY ANALYSES

[CKA under alternative matched-depth definitions; linear vs. RBF kernels.]

APPENDIX E: CODE AND DATA AVAILABILITY

[Repository, software versions, commands to reproduce every figure.]

BIBLIOGRAPHY

[1] Víctor Garcia Satorras, Emiel Hoogeboom, and Max Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*. PMLR, 2021.

[2] Samuel Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems*, 2019.

[3] Yi-Lun Liao and Tess Smidt. Equiformer: Equivariant graph attention transformer for 3d atomistic graphs. In *International Conference on Learning Representations*, 2023.

[4] Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, 2020. doi: 10.23915/distill.00024.001.

[5] Neel Nanda, Lawrence Chan, Tom Liberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023.

[6] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.

[7] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*. PMLR, 2019.

CURRICULUM VITAE

[Required. Full name, a contact address good for 1–2 years, education, positions, publications. Three to four pages at most.]